

KITABU QUEST

S1

MATHEMATICS



Cover scene

learners solving mathematics with charts and measuring tools with a mountain gorilla mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

1

Title Page

KITABU QUEST S1 Mathematics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S1 learners study Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use objects, drawings, number sentences, and worked examples before independent practice.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Sets
2. Sets of numbers
3. Linear functions, equations, and inequalities
4. Percentage, discount, profit, and loss
5. Ratio and proportions
6. Points, lines, and angles
7. Solids
8. Statistics (ungrouped data)
9. Probability

As you finish each topic, return to the success criteria and correct one answer before moving on.

Sets: Lesson Opener

Learning focus: Sets

Country link: a Rwandan school, market, farming, building, transport, or data problem for Sets.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- use sets, Venn diagrams, and relations to represent situations and solve problems
- define and give examples of sets
- indicate what a specified region in a Venn diagram represents, using connecting words (and, or, not) or set notation
- show how sets are used in representing given information
- observe a contextual problem that involves sets, record the solution using set notation, and give explanations
- demonstrate algebraic and graphical reasoning through the

Inquiry questions:

- How can sets help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Sets: Learn

Sets teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Sets
- method
- model
- unit
- answer
- check

Sets: Worked Example

Worked example for Sets: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Sets: Vocabulary And Study Skill

Use these words and study moves before you practise Sets. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Sets
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Sets without a clear example, method, or evidence.

Sets: Guided Activity

Guided activity for Sets:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Sets: Practice And Correction

Practice for Sets:

Main outcome: use sets, Venn diagrams, and relations to represent situations and solve problems.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to sets and solve it.

Sets: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Sets.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Sets: Outcome Check

Outcome check for Sets: Use sets, Venn diagrams, and relations to represent situations and solve problems.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Sets: Outcome Check

Outcome check for Sets: Define and give examples of sets.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Sets of numbers: Lesson Opener

Learning focus: Sets of numbers

Country link: a Rwandan school, market, farming, building, transport, or data problem for Sets of numbers.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- use operations to explore properties of sets of numbers and their relationships
- identify sets of numbers (natural, integer, rational, and real) and relationships between them
- illustrate different sets of numbers on a number line
- show that irrational numbers cannot be expressed exactly as a decimal
- carry out mathematical operations on sets of numbers
- work systematically to determine the operation properties of sets of numbers

Inquiry questions:

- How can sets of numbers help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Sets of numbers: Learn

Sets of numbers teaches numbers below, at, and above zero.

Core idea:

- Positive numbers are greater than zero.
- Negative numbers are less than zero.
- Zero is the turning point.
- A number line helps you compare and order integers.

Rwanda example: temperature changes, money owed, points gained or lost in a game, and height above or below a reference point can use integers.

Key words:

- Sets of numbers
- Sets
- numbers
- method
- model
- unit
- answer
- check

Sets of numbers: Worked Example

Worked example for Sets of numbers: The morning temperature is 3 degrees above zero. At night it drops by 7 degrees. What is the night temperature?

1. Start at +3 on a number line.
2. Move 7 steps left because the temperature drops.
3. $+3 - 7 = -4$.

Answer: 4 degrees below zero.

Sets of numbers: Vocabulary And Study Skill

Use these words and study moves before you practise Sets of numbers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Sets of numbers
- Sets
- numbers
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Sets of numbers without a clear example, method, or evidence.

Sets of numbers: Guided Activity

Guided activity for Sets of numbers:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Sets of numbers: Practice And Correction

Practice for Sets of numbers:

Main outcome: use operations to explore properties of sets of numbers and their relationships.

1. Solve one sets of numbers question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to sets of numbers and solve it.

Sets of numbers: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Sets of numbers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Sets of numbers: Outcome Check

Outcome check for Sets of numbers: Use operations to explore properties of sets of numbers and their relationships.

1. Place -3, 0, and +4 on a number line.
2. Which number is greatest?
3. Find the distance from -3 to +4.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Sets of numbers: Outcome Check

Outcome check for Sets of numbers: Identify sets of numbers (natural, integer, rational, and real) and relationships between them.

1. Place -3, 0, and +4 on a number line.
2. Which number is greatest?
3. Find the distance from -3 to +4.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Linear functions, equations, and inequalities: Lesson Opener

Learning focus: Linear functions, equations, and inequalities

Country link: a Rwandan school, market, farming, building, transport, or data problem for Linear functions, equations, and inequalities.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- represent and interpret graphs of linear functions and apply them in real life situations; solve linear equations and inequalities; appreciate the importance of checking solutions; and represent the solution
- define a linear function and recognize its graph
- illustrate that a linear function is written in the form $y = mx + c$, where c is the y-intercept, m is a measure of steepness, and the solution of the equation is the x-intercept
- explain what is meant by the solution of a linear equation and inequality
- plot linear functions on the Cartesian plane
- interpret the graph of a linear function linking the parameters of the function with the features of the graph, including intercepts and steepness

Inquiry questions:

- How can linear functions, equations, and inequalities help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Linear functions, equations, and inequalities: Learn

Linear functions, equations, and inequalities teaches how lines and angles describe shapes and directions.

Core idea:

- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:

- Linear functions, equations, and inequalities
- Linear
- functions
- equations
- inequalities
- method
- model
- unit

Linear functions, equations, and inequalities: Worked Example

Worked example for Linear functions, equations, and inequalities: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Linear functions, equations, and inequalities: Vocabulary And Study Skill

Use these words and study moves before you practise Linear functions, equations, and inequalities. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Linear functions, equations, and inequalities
- Linear
- functions
- equations
- inequalities
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Linear functions, equations, and inequalities without a clear example, method, or evidence.

Linear functions, equations, and inequalities: Guided Activity

Guided activity for Linear functions, equations, and inequalities:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Linear functions, equations, and inequalities: Practice And Correction

Practice for Linear functions, equations, and inequalities:

Main outcome: represent and interpret graphs of linear functions and apply them in real life situations; solve linear equations and inequalities; appreciate the importance of checking solutions; and represent the solution.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:

Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to linear functions, equations, and inequalities and solve it.

Linear functions, equations, and inequalities: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Linear functions, equations, and inequalities.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Linear functions, equations, and inequalities: Outcome Check

Outcome check for Linear functions, equations, and inequalities: Represent and interpret graphs of linear functions and apply them in real life situations; solve linear equations and inequalities; appreciate the importance of checking solutions; and represent the solution.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Linear functions, equations, and inequalities: Outcome Check

Outcome check for Linear functions, equations, and inequalities: Define a linear function and recognize its graph.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Percentage, discount, profit, and loss: Lesson Opener

Learning focus: Percentage, discount, profit, and loss

Country link: a Rwandan school, market, farming, building, transport, or data problem for Percentage, discount, profit, and loss.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve problems that involves calculating percentage, discount, profit, loss, and other financial calculations
- explain how to calculate discount, commission, profit and loss, simple interest, tax
- use percentages to calculate discount, commission, profit, loss, interest, taxes
- solve problems involving: Discount, commission, profit, loss, loans, savings, tax and insurance
- appreciate the role money plays in our lives
- be honest in managing and using money

Inquiry questions:

- How can percentage, discount, profit, and loss help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Percentage, discount, profit, and loss: Learn

Percentage, discount, profit, and loss teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:

- Percentage, discount, profit, and loss
- Percentage
- discount
- profit
- loss
- method
- model
- unit

Percentage, discount, profit, and loss: Worked Example

Worked example for Percentage, discount, profit, and loss: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:

1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Percentage, discount, profit, and loss: Vocabulary And Study Skill

Use these words and study moves before you practise Percentage, discount, profit, and loss. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Percentage, discount, profit, and loss
- Percentage
- discount
- profit
- loss
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Percentage, discount, profit, and loss without a clear example, method, or evidence.

Percentage, discount, profit, and loss: Guided Activity

Guided activity for Percentage, discount, profit, and loss:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Percentage, discount, profit, and loss: Practice And Correction

Practice for Percentage, discount, profit, and loss:

Main outcome: solve problems that involves calculating percentage, discount, profit, loss, and other financial calculations.

1. Solve one percentage, discount, profit, and loss question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to percentage, discount, profit, and loss and solve it.

Percentage, discount, profit, and loss: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Percentage, discount, profit, and loss.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Percentage, discount, profit, and loss: Outcome Check

Outcome check for Percentage, discount, profit, and loss: Solve problems that involves calculating percentage, discount, profit, loss, and other financial calculations.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Percentage, discount, profit, and loss: Outcome Check

Outcome check for Percentage, discount, profit, and loss: Explain how to calculate discount, commission, profit and loss, simple interest, tax.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Ratio and proportions: Lesson Opener

Learning focus: Ratio and proportions

Country link: a Rwandan school, market, farming, building, transport, or data problem for Ratio and proportions.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve problems involving ratio and proportion
- express ratios in their simplest form
- identify a direct and indirect proportion
- differentiate direct from indirect proportion
- compare quantities using proportions
- share quantities in a given proportion or ratio

Inquiry questions:

- How can ratio and proportions help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Ratio and proportions: Learn

Ratio and proportions teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Ratio and proportions
- Ratio
- proportions
- method
- model
- unit
- answer
- check

Ratio and proportions: Worked Example

Worked example for Ratio and proportions: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Ratio and proportions: Vocabulary And Study Skill

Use these words and study moves before you practise Ratio and proportions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Ratio and proportions
- Ratio
- proportions
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Ratio and proportions without a clear example, method, or evidence.

Ratio and proportions: Guided Activity

Guided activity for Ratio and proportions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Ratio and proportions: Practice And Correction

Practice for Ratio and proportions:

Main outcome: solve problems involving ratio and proportion.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to ratio and proportions and solve it.

Ratio and proportions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Ratio and proportions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Ratio and proportions: Outcome Check

Outcome check for Ratio and proportions: Solve problems involving ratio and proportion.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Ratio and proportions: Outcome Check

Outcome check for Ratio and proportions: Express ratios in their simplest form.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Points, lines, and angles: Lesson Opener

Learning focus: Points, lines, and angles

Country link: a Rwandan school, market, farming, building, transport, or data problem for Points, lines, and angles.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- to be able to construct mathematical arguments using the angle properties of parallel lines
- recognize that the position of an angle at a point sum to 360° ; angles at a point on a straight line sum to 180°
- distinguish and recognize vertically opposite, corresponding, alternate, and supplementary angles
- use knowledge of angle properties of parallel lines and shapes to construct arguments when finding missing angles in geometric diagrams
- construct and calculate angles
- appreciate the need to give reasons when developing solutions to missing angle problems

Inquiry questions:

- How can points, lines, and angles help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Points, lines, and angles: Learn

Points, lines, and angles teaches how lines and angles describe shapes and directions.

Core idea:

- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:

- Points, lines, and angles
- Points
- lines
- angles
- method
- model
- unit
- answer

Points, lines, and angles: Worked Example

Worked example for Points, lines, and angles: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Points, lines, and angles: Vocabulary And Study Skill

Use these words and study moves before you practise Points, lines, and angles. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Points, lines, and angles
- Points
- lines
- angles
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Points, lines, and angles without a clear example, method, or evidence.

Points, lines, and angles: Guided Activity

Guided activity for Points, lines, and angles:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Points, lines, and angles: Practice And Correction

Practice for Points, lines, and angles:

Main outcome: to be able to construct mathematical arguments using the angle properties of parallel lines.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:

Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to points, lines, and angles and solve it.

Points, lines, and angles: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Points, lines, and angles.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Points, lines, and angles: Outcome Check

Outcome check for Points, lines, and angles: To be able to construct mathematical arguments using the angle properties of parallel lines.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Points, lines, and angles: Outcome Check

Outcome check for Points, lines, and angles: Recognize that the position of an angle at a point sum to 360°; angles at a point on a straight line sum to 180°.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Solids: Lesson Opener

Learning focus: Solids

Country link: a Rwandan school, market, farming, building, transport, or data problem for Solids.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- select and use formulae to find the surface area and volume of solids
- explain the surface area of a solid as the area of the net
- illustrate the volume as the space occupied by a solid
- distinguish between surface area and volume and know the correct units
- derive the surface area for prisms and cylinders
- calculate the surface area and volume of common geometrical solids using formulas where necessary

Inquiry questions:

- How can solids help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Solids: Learn

Solids teaches properties of three-dimensional shapes such as cubes, cuboids, and cylinders.

Core idea:

- Solid shapes have faces, edges, and vertices.
- A cuboid has rectangular faces.
- A cube has six equal square faces.
- A cylinder has two circular faces and one curved surface.
- Nets show how flat faces fold to make a solid.

Key words:

- Solids
- method
- model
- unit
- answer
- check

Solids: Worked Example

Worked example for Solids: Name the faces of a cuboid.

1. Look at the solid shape.
2. A cuboid has 6 faces.
3. Each face is a rectangle.
4. Opposite faces are equal in size.

Answer: A cuboid has 6 rectangular faces, 12 edges, and 8 vertices.

Solids: Vocabulary And Study Skill

Use these words and study moves before you practise Solids. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Solids
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Solids without a clear example, method, or evidence.

Solids: Guided Activity

Guided activity for Solids:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Solids: Practice And Correction

Practice for Solids:

Main outcome: select and use formulae to find the surface area and volume of solids.

1. Draw or model a cube, cuboid, and cylinder.
2. Count faces, edges, and vertices where they apply.
3. Sketch a simple net for a cube or cuboid.

Answer check:

Answer check: cube has 6 equal square faces; cuboid has 6 rectangular faces; cylinder has circular faces and a curved surface.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to solids and solve it.

Solids: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Solids.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Solids: Outcome Check

Outcome check for Solids: Select and use formulae to find the surface area and volume of solids.

1. Name the faces of a cuboid.
2. Count edges and vertices of a cube.
3. Describe one difference between a cuboid and a cylinder.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Solids: Outcome Check

Outcome check for Solids: Explain the surface area of a solid as the area of the net.

1. Name the faces of a cuboid.
2. Count edges and vertices of a cube.
3. Describe one difference between a cuboid and a cylinder.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics (ungrouped data): Lesson Opener

Learning focus: Statistics (ungrouped data)

Country link: a Rwandan school, market, farming, building, transport, or data problem for Statistics (ungrouped data).

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- to be able to collect, represent, and interpret quantitative discrete data appropriate to a question or problem
- define quantitative data and qualitative data
- differentiate discrete and continuous data
- present data on a frequency distribution
- define mode and median of given statistical data
- recognize formulae used to calculate the mean and median

Inquiry questions:

- How can statistics (ungrouped data) help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics (ungrouped data): Learn

Statistics (ungrouped data) teaches how to collect, organize, read, and explain information.

Core idea:

- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:

- Statistics (ungrouped data)
- Statistics
- ungrouped
- data
- method
- model
- unit
- answer

Statistics (ungrouped data): Worked Example

Worked example for Statistics (ungrouped data): A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Statistics (ungrouped data): Vocabulary And Study Skill

Use these words and study moves before you practise Statistics (ungrouped data). Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Statistics (ungrouped data)
- Statistics
- ungrouped
- data
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Statistics (ungrouped data) without a clear example, method, or evidence.

Statistics (ungrouped data): Guided Activity

Guided activity for Statistics (ungrouped data):

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Statistics (ungrouped data): Practice And Correction

Practice for Statistics (ungrouped data):

Main outcome: to be able to collect, represent, and interpret quantitative discrete data appropriate to a question or problem.

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:

Answer check: totals in the tally table and bar chart should match.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to statistics (ungrouped data) and solve it.

Statistics (ungrouped data): Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Statistics (ungrouped data).
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics (ungrouped data): Outcome Check

Outcome check for Statistics (ungrouped data): To be able to collect, represent, and interpret quantitative discrete data appropriate to a question or problem.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Statistics (ungrouped data): Outcome Check

Outcome check for Statistics (ungrouped data): Define quantitative data and qualitative data.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Probability: Lesson Opener

Learning focus: Probability

Country link: a Rwandan school, market, farming, building, transport, or data problem for Probability.

Progression focus: build lower-secondary foundations with clear vocabulary and guided application; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- to be able to determine the probability of an event happening using equally likely events or experiments
- define an event and explain why probabilities can only be between 0 (impossible) and 1 (certain)
- explain that probabilities can be calculated using equally likely outcomes (e.g. tossing a coin or dice, drawing a card from a deck) or estimated using experimental data (e.g. weather, sports, arriving late to school)
- demonstrate that the more data collected, the better the estimate of the probability
- calculate the probability of an event where there are equally likely outcomes e.g. heads or tails on a coin, a score on a dice
- estimate probabilities using data

Inquiry questions:

- How can probability help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Probability: Learn

Probability teaches how likely an event is.

Core idea:

- Some events are certain, likely, unlikely, or impossible.
- A simple experiment can help you compare chances.
- Record results before making a conclusion.

Key words:

- Probability
- method
- model
- unit

Probability: Worked Example

Worked example for Probability: A bag has 4 red counters and 1 blue counter. Which colour is more likely to be picked?

1. Compare the number of counters.
2. Red has 4 chances; blue has 1 chance.
3. More red counters means red is more likely.

Answer: Red is more likely.

Probability: Vocabulary And Study Skill

Use these words and study moves before you practise Probability. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Probability
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Probability without a clear example, method, or evidence.

Probability: Guided Activity

Guided activity for Probability:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Probability: Practice And Correction

Practice for Probability:

Main outcome: to be able to determine the probability of an event happening using equally likely events or experiments.

1. Write one certain event, one likely event, and one impossible event.
2. Toss a coin 10 times and record heads/tails.
3. Use your results to say what happened more often.

Answer check:

Answer check: certain means must happen; impossible means cannot happen.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to probability and solve it.

Probability: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Probability.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Probability: Outcome Check

Outcome check for Probability: To be able to determine the probability of an event happening using equally likely events or experiments.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Probability: Outcome Check

Outcome check for Probability: Define an event and explain why probabilities can only be between 0 (impossible) and 1 (certain).

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Sets: Review Clinic 1

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Sets that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Sets of numbers: Review Clinic 2

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Sets of numbers that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Linear functions, equations, and inequalities: Review Clinic 3

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Linear functions, equations, and inequalities that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Percentage, discount, profit, and loss: Review Clinic 4

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Percentage, discount, profit, and loss that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Mathematics.

- Sets: Explain this term using an example from the book, your school, or your community.
- Sets of numbers: Explain this term using an example from the book, your school, or your community.
- Linear functions, equations, and inequalities: Explain this term using an example from the book, your school, or your community.
- Percentage, discount, profit, and loss: Explain this term using an example from the book, your school, or your community.
- Ratio and proportions: Explain this term using an example from the book, your school, or your community.
- Points, lines, and angles: Explain this term using an example from the book, your school, or your community.
- Solids: Explain this term using an example from the book, your school, or your community.
- Statistics (ungrouped data): Explain this term using an example from the book, your school, or your community.
- Probability: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Mathematics that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.